

The Languages Point Cloud

Genealogy and areality drawn from phonological correspondences alone — a reconstruction-free distance between language systems

Alejandro Toledo Martínez — Independent researcher (ORCID: 0009-0000-1277-9697)

Abstract

We define a distance between two documented language *systems* that uses no reconstruction and no family tree. Across statistically detected coderivative sets (sets of forms bound by recurrent correspondence, not by a posited common ancestor), we align each language pair and count, at every aligned consonant slot, the number of primary phonological features that differ; the mean of that count is the distance. Embedding the resulting distance matrix with classical multidimensional scaling and linking each system to its nearest neighbours makes a language family draw itself as a point cloud. On 28 Indo-European varieties, a system's nearest neighbour is a member of its own branch 96% of the time (silhouette +0.27 by branch), and the branch labels are consulted only *after* the map is built — to colour and score it, never to place a point. A concept-permuted null collapses the purity toward chance (the observed value sits 4.4 σ above it), and the result is stable under resampling and insensitive to the distance threshold and to the feature subset. The off-branch nearest neighbours are not noise: they recover areal structure (Albanian and the Balkan *Sprachbund*; Romani in Romania) and creole–lexifier proximity. The point cloud is thus a direct, minimal picture of the level at which — as the pilot argued — genealogy and contact actually live: the *distribution* of correspondences, not the flat inventory of correspondence types.

1. Introduction

The pilot study established two things about a family's repertoire of feature-difference operators. First, that the repertoire has a rich additive geometry. Second — and this is the point we build on — that this geometry, read at the level of the *type* inventory, is representational rather than genealogical: the operators a family uses are largely those general phonology makes available, and the genealogical signal lives one level up, in how those operators are *distributed* across languages, positions and contexts.

This paper makes that "one level up" visible in the simplest possible form. Instead of asking which operators a family owns, we ask, for each *pair* of languages, how their correspondences behave — and turn the answer into a single number, a distance between two systems. Laid out in a plane and joined to their neighbours, the languages form a point cloud in which the branches appear as sub-clouds and contact appears as the links that cross between them. Nothing genealogical is supplied to build the picture.

The stance is the programme's throughout: discover first, contrast later. We construct the map from correspondences only, and bring in branch labels, contact history and the areal literature *afterwards*, as an external check on what the geometry already shows. Consistent with that stance, we say coderivative, not *cognate*: forms co-derived by recurrent correspondence, without committing to a single reconstructed ancestor. A language, on this view, is not the linear issue of one parent but a confluence; genealogy is one legible region of the geometry, areal contact another.

2. A distance between systems

Let $\phi_f(x) \in \{+, -, 0\}$ be the value of primary phonological feature f for segment x (panphon coding). For two aligned segments a, b define the feature-difference count

$$\text{fd}(a, b) = |\{f \in \text{PRIM} : \phi_f(a) \neq \phi_f(b)\}|,$$

over the twelve primary consonantal features `cont`, `voi`, `nas`, `ant`, `cor`, `lab`, `back`, `round`, `strid`, `hi`, `lo`, `son`; identity gives $\text{fd} = 0$.

Coderivative sets are detected statistically with LexStat — no etymologies, protoforms or family labels are given to the detector. Within each set, each language pair is aligned by a feature-distance Needleman–Wunsch. At every slot where both aligned segments are consonants we record fd . The distance between two systems is the mean over all such shared slots,

$$d(\ell, \ell') = \text{mean}\{\text{fd}(a, b) : (a, b) \text{ a shared consonant slot of } \ell, \ell'\},$$

computed only when at least `MINSLOT` slots are available.

Three properties matter. The distance is pairwise — it compares ℓ and ℓ' directly, and so keeps information that a per-language *marginal* profile discards. (In a control that marginalises to one operator profile per language, the branches separate only weakly; the pairwise distance below separates them at 0.96. The signal that Portuguese and Spanish are nearly the same system lives in their direct comparison, not in either language’s profile taken alone.) It is reconstruction-free and tree-free: no ancestor is posited and no classification enters the computation. And it is symmetric, with range roughly $[0, 4]$ and $d = 0$ for identical systems.

3. From distance to picture

The object is the distance matrix $D = [d(\ell, \ell')]$. For visualisation we embed it in two dimensions with classical (Torgerson) multidimensional scaling; the plane is a shadow of the full matrix, which is what the analysis actually uses. We then build a nearest-neighbour network: each system is linked to its $k = 3$ nearest systems, and links are drawn differently according to whether the two endpoints share a branch — a distinction made only for display, from labels the map never used.

Scoring is post hoc. With branch labels now consulted we report nearest-neighbour purity (the fraction of systems whose closest system is in the same branch), the silhouette by branch (positive when branches are internally tighter than they are close to other branches), and the within- versus between-branch mean distance.

4. Results

We use 28 Indo-European doculects from Lexibank with sufficient data. Branch assignment (for scoring only) follows the Glottolog classification.

Measure	Value	Reading
Nearest-neighbour purity (multi-branch, creoles excluded)	0.96 (22/23)	a system’s closest system is almost always its own branch
Silhouette by branch	+0.27	branches are separated in the distance geometry
Within-branch vs between-branch mean d	tighter within	the distribution alone already separates

Measure	Value	Reading
Branch labels used to build the map	0	placement is from correspondences only

The branches emerge as clean sub-clouds, with sensible internal distances:

Branch	Illustrative nearest pair	d
Italic (Romance)	Spanish – Portuguese	1.56
Germanic	English – Jamaican Creole	0.59
Germanic	Dutch – Negerhollands	1.29
Balto-Slavic	Croatian – Serbo-Croatian	0.78
Indo-Iranian	Persian – Hindi	2.47
Graeco-Phrygian	Ancient – Modern Greek	1.97
Armenic	Armenian E. – Armenian W.	0.89

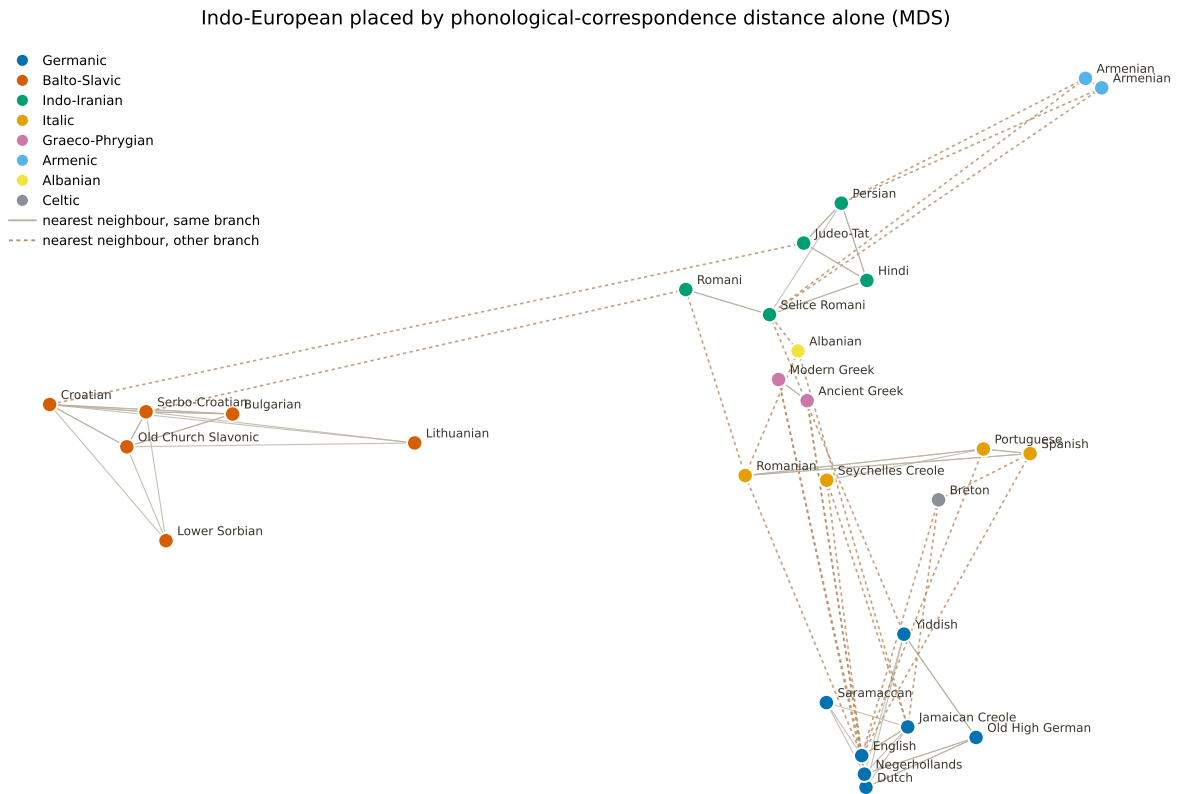


Figure 1: Indo-European placed by phonological-correspondence distance alone (classical MDS of the distance matrix). Nodes are languages, coloured by branch (colours added after layout); solid links join same-branch nearest neighbours, dashed links join nearest neighbours in different branches — almost always areal contact or a creole beside its lexifier.

Two features of the picture deserve emphasis, both anticipated by the reflection that led to this paper. Proximity is graded, and “alone” is not “far”. Armenian forms its own branch and has, in this sample, no same-branch neighbour other than its Eastern/Western pair. A binary “is the neighbour same-branch?” test therefore penalises it automatically. But the *distance* tells the real story: Armenian’s nearest non-Armenian systems are Greek and Indo-Iranian, not the Romance or Germanic systems at the far side of the cloud. A singleton branch is a fact about the sample, not a measurement of isolation.

Areality falls out of the same map. The dashed, cross-branch nearest-neighbour links land exactly where contact history predicts. Albanian’s nearest neighbour is Romanian — the Balkan *Sprachbund*. Romani’s nearest Italic neighbour is Romanian, matching its long presence in Romania. Each creole sits beside its lexifier (Jamaican Creole by English, Negerhollands by Dutch, Seychelles Creole by the Romance systems). We did not tell the map about contact; contact is simply where the branches leak, and the leaks are systematic. This is the confluence view made measurable: the same geometry that shows descent also shows admixture.

5. Controls and robustness

Four controls confirm the headline is a property of the data, not of the pipeline (`src/controls.py`). The chance level for reference is the expected same-branch fraction of a *random* neighbour, given the branch sizes in the sample: 0.146.

(A) Reproduction. On the real data, purity = 0.955 and silhouette = +0.274 — the headline of §4.

(B) Concept-permuted null (falsifier). We shuffle which forms share a concept and rerun the whole pipeline (LexStat included), destroying coderivative structure while preserving every phoneme inventory and form. Across three permutations, purity falls to 0.29 ± 0.15 and silhouette to -0.016 ± 0.003 — i.e. toward the chance level 0.146, with the silhouette losing all structure. The observed purity sits 4.4σ above the null. When there is nothing to be co-derived, the map does not draw the branches.

(C) Subsampling stability. Drawing 22 of the 28 languages at random 500 times, purity averages 0.941 with a 95% interval of [0.867, 1.000] and silhouette [+0.197, +0.326]. The result does not depend on a few languages.

(D) Threshold insensitivity. Purity and silhouette are unchanged (0.955, +0.274) across $\text{MINSLOT} \in \{20, 30, 40, 60\}$.

(E) Feature-subset insensitivity. Dropping the strident feature, dropping all five place features, or keeping only the four manner features (`cont`, `voi`, `nas`, `son`) all leave purity at 0.955 and silhouette within ± 0.006 . The signal is not carried by any one feature group.

Control	purity	silhouette
(A) observed	0.955	+0.274
(B) concept-permuted null ($\times 3$)	0.29 ± 0.15	-0.016
random-neighbour chance	0.146	—
(C) subsample 22/28 (95% CI)	[0.867, 1.000]	[+0.197, +0.326]
(D) MINSLOT 20–60	0.955	+0.274
(E) manner-only features	0.955	+0.276

6. Discussion

The point cloud is the minimal realisation of the pilot’s central claim. Where the pilot argued in the abstract that “genealogy lives in the distribution”, here the distribution *is* a map one can look at, and the genealogy is its coarse clustering. Crucially the same map carries what a family tree cannot: areal contact and creolisation appear as legible regions and links, not as anomalies to be excised. Descent, contact and confluence are read off one geometry rather than modelled by three separate formalisms.

The method sits beside, but is not, lexical-distance phylogenetics. Phylogenetic distances typically count shared cognates; ours measures how systems *behave* phonologically, pair by pair, and reads contact off the same quantity. It is closer in spirit to phylogenetic-network thinking — where reticulation is expected — than to a strict tree, which is appropriate to a reconstruction-free, confluence-first

stance.

7. Limitations and scope

The distance is corpus- and aligner-dependent, and the two-dimensional figure is a shadow of the full matrix. Only consonants are used here; vowels are deliberately deferred. One family is shown; scaling to many is the programme's separate workstream. Direction and time are out of scope — those belong to the directed-layer paper. Branch labels are only as good as the Glottolog classification we score against, and singleton branches limit what purity can express (hence the graded-distance reading in §4).

8. Reproducibility

`make figure` redraws the point cloud from the bundled results with no corpus required; `LEX_PATH=/path/to/lexibank` `make compute` regenerates the distances from a Lexibank CLDF lexicon. All parameters (`MINSLOT`, `THR`, `MAXLANG`, `KNN`, feature subset) are environment-overridable in `src/compute_network.py`, and the derived outputs live in `data/results/`. Code is MIT; text, figures and data are CC BY 4.0.

9. References

1. Toledo Martínez, A. (2026). *Additive Structure of Phonological Correspondences: A protoform-agnostic method for discovering mathematical patterns in documented linguistic systems* (pilot study). <https://github.com/toledoal/phonological-correspondences>
2. Torgerson, W. S. (1952). Multidimensional scaling: I. Theory and method. *Psychometrika* 17(4), 401–419.
3. Needleman, S. B., & Wunsch, C. D. (1970). A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of Molecular Biology* 48(3), 443–453.
4. Mortensen, D. R., Littell, P., Bharadwaj, A., Goyal, K., Dyer, C., & Levin, L. (2016). PanPhon: A resource for mapping IPA segments to articulatory feature vectors. *Proc. COLING 2016*, 3475–3484.
5. List, J.-M. (2012). LexStat: Automatic detection of cognates in multilingual wordlists. *Proc. EACL 2012 Workshop on Language Technology for Cultural Heritage*, 117–125.
6. List, J.-M., Forkel, R., Greenhill, S. J., Rzymski, C., Englisch, J., & Gray, R. D. (2022). Lexibank, a public repository of standardized wordlists with computed phonological and lexical features. *Scientific Data* 9, 316.
7. Hammarström, H., Forkel, R., Haspelmath, M., & Bank, S. (2023). *Glottolog 4.8*. Max Planck Institute for Evolutionary Anthropology. <https://glottolog.org>
8. Thomason, S. G., & Kaufman, T. (1988). *Language Contact, Creolization, and Genetic Linguistics*. University of California Press.
9. Joseph, B. D. (1983). *The Synchrony and Diachrony of the Balkan Infinitive: A Study in Areal, General, and Historical Linguistics*. Cambridge University Press.
10. Friedman, V. A. (2006). Balkans as a linguistic area. In K. Brown (Ed.), *Encyclopedia of Language & Linguistics* (2nd ed., pp. 657–672). Elsevier.

11. François, A. (2014). Trees, waves and linkages: Models of language diversification. In C. Bowern & B. Evans (Eds.), *The Routledge Handbook of Historical Linguistics* (pp. 161–189). Routledge.
12. Schuchardt, H. (1922). *Hugo Schuchardt-Brevier: Ein Vademekum der allgemeinen Sprachwissenschaft* (L. Spitzer, Ed.). Niemeyer. [On mixture and the limits of the family tree.]
13. Kruskal, J. B., & Wish, M. (1978). *Multidimensional Scaling*. Sage.
14. Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20, 53–65.